

COMPATIBILITY OF SPECTRAL AND OPERATOR REDUCTIONS OF A FIRST-PRINCIPLES SILICON HAMILTONIAN

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ABSTRACT

First-principles electronic Hamiltonians may be reduced to compact lattice models by fitting selected eigenvalues or by approximating an aligned localized operator. These criteria need not select the same effective Hamiltonian. This study formulates their compatibility within a common hierarchy of orthogonal sp^3s^* Slater–Koster models for bulk silicon. A validated ten-orbital Wannier Hamiltonian will be constructed from one converged density-functional-theory calculation. For each tight-binding model class, the spectral and operator criteria define admissible sets satisfying declared error tolerances. A normalized real-space Hamiltonian metric will determine the minimum separation between these sets. A vanishing separation identifies a common Hamiltonian satisfying both requirements; a nonzero separation measures the incompatibility of the prescribed model class with the retained first-principles information. Validation will use withheld band energies, the indirect gap, conduction-valley position, longitudinal and transverse electron effective masses, and operator residuals resolved by orbital block and neighbor shell. The study will identify the smallest tested model class providing a common endpoint for the two reductions and thereby establish a validated bulk reference for subsequent impurity-operator reduction.

KEYWORDS: silicon; Wannier Hamiltonian; tight-binding reduction; operator compatibility; model selection

INTRODUCTION

Density-functional theory (DFT) supplies a first-principles Kohn–Sham description of crystalline electronic structure, but its plane-wave representation is inconvenient for interpolation and reduced lattice calculations. Maximally localized Wannier functions provide a localized representation of a selected Kohn–Sham subspace and support accurate interpolation [5], [7], [9], [12]. Parameterized tight-binding models instead restrict the Hamiltonian to a prescribed orbital basis, hopping range, and symmetry structure [8], [11].

A spectral fit preserves selected eigenvalues, whereas an operator fit preserves matrix information in an aligned localized representation. Spectral agreement alone does not establish agreement of onsite terms, orbital couplings, or hopping amplitudes. These differences are consequential when the bulk Hamiltonian is later modified by an impurity or another local perturbation. Before an operator residual can be evaluated, the Wannier and tight-binding orbital coordinates must therefore be identified through a stable, symmetry-compatible alignment [1], [4].

The present study asks whether the spectral and operator criteria admit a common tight-binding Hamiltonian within declared tolerances. Rather than forcing two independently fitted parameter vectors to agree, it measures the distance between the sets of models accepted by each criterion. Repeating this test over a nested hierarchy of tight-binding model classes identifies the smallest tested class for which the two admissible sets intersect, or quantifies the obstruction when they do not.

METHODOLOGY

A converged bulk-silicon reference will be calculated with QUANTUM ESPRESSO[2] using PBE-GGA [6], a scalar-relativistic PseudoDojo optimized norm-conserving pseudopotential [3], [10], the PBE-relaxed lattice constant, and a non-spin-polarized, non-SOC parent problem. Plane-wave cutoffs, wavevector meshes, and the number of bands will be converged against the indirect gap, conduction-valley position, and longitudinal and transverse electron effective masses.

A ten-orbital target subspace containing the valence bands and the low conduction states required for the silicon valleys will then be constructed with Wannier90 [7], [9]. Its real-space Hamiltonian is

$$[\mathbf{H}_W(\mathbf{R})]_{\alpha\beta} = \langle w_{\alpha 0} | \hat{H}_{KS} | w_{\beta \mathbf{R}} \rangle,$$

where $\mathbf{R}$ is a direct-lattice translation; α and β label Wannier orbitals in the reference and translated primitive cells, respectively; $|w_{\alpha 0}\rangle$ and $|w_{\beta \mathbf{R}}\rangle$ are localized Wannier states; and $\hat{H}_{KS}$ is the converged Kohn–Sham operator.

Initial projections and outer and frozen disentanglement windows will be recorded. The Wannier Hamiltonian will be accepted as the common parent only after its dense-grid interpolation error, centers, spreads, sensitivity to window and projection choices, and real-space hopping decay have been validated.

For each candidate model class, the spectral loss is a weighted error over training-set Wannier eigenvalues and selected band-edge quantities. The operator loss is

$$\mathcal{L}_H(\mathbf{C}, \boldsymbol{\theta}) = \sum_{\mathbf{R}} \omega_{\mathbf{R}} \left\| \mathbf{H}_W(\mathbf{R}) - \mathbf{C} \mathbf{H}_{\text{TB}}(\mathbf{R}; \boldsymbol{\theta}) \mathbf{C}^\dagger \right\|_F^2,$$

where $\mathbf{C} \in \mathcal{U}$ is a symmetry-compatible unitary alignment from the tight-binding orbital coordinates to the Wannier coordinates; $\boldsymbol{\theta}$ is the tight-binding parameter vector; $\mathbf{H}_{\text{TB}}(\mathbf{R}; \boldsymbol{\theta})$ is the tight-binding hopping matrix associated with $\mathbf{R}$; $\omega_{\mathbf{R}} \geq 0$ weights the contribution of each retained neighbor shell; and $\|\cdot\|_F$ is the Frobenius norm.

The model hierarchy will begin with an orthogonal ten-orbital sp^3s^* Hamiltonian with nearest-neighbor hopping. If that class is incompatible with the declared tolerances, second-neighbor and selected symmetry-allowed corrections will be introduced in a fixed sequence.

For a candidate model class $\mathfrak{M}_j$, the two losses define the admissible sets

$$\begin{aligned} \mathfrak{A}_{E,j}(\tau_E) &= \{\mathbf{H}(\boldsymbol{\theta}) \in \mathfrak{M}_j : \mathcal{L}_E(\boldsymbol{\theta}) \leq \tau_E\}, \\ \mathfrak{A}_{H,j}(\tau_H) &= \left\{ \mathbf{H}(\boldsymbol{\theta}) \in \mathfrak{M}_j : \min_{\mathbf{C} \in \mathcal{U}} \mathcal{L}_H(\mathbf{C}, \boldsymbol{\theta}) \leq \tau_H \right\}, \end{aligned}$$

where: $\mathcal{L}_E$ is the spectral loss; $\mathcal{L}_H$ is the aligned operator loss; τ_E is the declared spectral tolerance; τ_H is the declared operator tolerance; and j indexes the nested tight-binding model classes.

The spectral tolerance includes targets of 5% for the indirect gap, 3% of the Γ – X distance for the conduction-valley position, and 8% for each electron effective mass. Operator and alignment tolerances will be normalized to the converged Wannier Hamiltonian and fixed before the compatibility search.

A dimensionless real-space Hamiltonian metric d_{TB} will compare models in their common canonical tight-binding coordinates. The minimum distance between the two admissible sets is

$$\delta_j^* = \inf_{\substack{\mathbf{H}_E \in \mathfrak{A}_{E,j} \\ \mathbf{H}_H \in \mathfrak{A}_{H,j}}} d_{\text{TB}}(\mathbf{H}_E, \mathbf{H}_H),$$

where $\mathbf{H}_E$ and $\mathbf{H}_H$ are models satisfying the spectral and operator criteria, respectively. A value of δ_j^* below the declared map tolerance establishes compatibility within $\mathfrak{M}_j$; otherwise, it quantifies the irreducible separation under the imposed spectral and operator requirements.

All accepted models will be tested on withheld wavevectors. The operator residual will be decomposed by onsite and hopping contribution, orbital block, crystal-symmetry channel, and neighbor shell. The analysis is restricted to bulk silicon; impurity operators are outside the present study.

EXPECTED RESULT

The required outputs are one converged bulk-silicon DFT dataset, one validated ten-orbital Wannier Hamiltonian, admissible spectral and operator model sets for each tested tight-binding

class, their minimum metric separation, and one common validation record. Numerical entries will be reported only after the econvergence, fitting, and sensitivity calculations are complete.

Quantity	Mathematical criterion	Decision role
Spectral admissibility	Spectral loss within threshold	Retains selected bands and band-edge quantities
Operator admissibility	Aligned operator loss within threshold	Retains real-space matrix information
Map compatibility	Set distance within map threshold	Establishes a common reduced Hamiltonian
Model selection	Lowest-order compatible class	Selects the minimal common Hamiltonian class

The decisive result is the smallest tested model class for which the spectral and operator admissible sets intersect within tolerance. If no tested class is compatible, the nonzero set separation and its decomposition will identify the orbital or hopping content responsible for the obstruction. Either outcome provides a controlled basis for deciding which bulk Hamiltonian may be modified in subsequent impurity calculations. Construction of doped-supercell and impurity operators remains outside the present scope.

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